

DRAFT FOR REVIEW

Robo-K9

Robot scent-detection dog: bed bug proof of concept

Research, build plan, bill of materials, schematics, robot link-up and hub

Date	September 25, 2026
Version	0.3 (adds the robot dog link-up and the hub)
Target budget	≈ \$3,600–4,100 recommended path; ≈ \$600–800 for the Phase 0 sensor bench
Timeline	~24 weeks to field pilot, plus an optional Phase 5 R&D track
Source	github.com/turkokami/barber-shack · robot-scent-dog/

Contents

1. Executive summary and questions for reviewers
2. Build plan: research, architecture, phases, BOM, budget, validation, safety
3. Schematics: system, pneumatic, electrical, power, mechanical
4. Robot dog link-up and hub: Go2 Pro / EDU link-up schematics, duties, reporting

Prices are approximate as of September 2026; check each purchase link at checkout. Firmware in the repository has been checked on simulated data only and has not run on hardware yet.

Executive summary

A quadruped robot with an electronic nose ("e-nose") that walks a room, sniffs likely hiding spots, and **sits** when it smells bed bugs, the way a trained detection dog alerts.

One-paragraph summary

Bed bugs give off a known chemical signature: the alarm pheromone **(E)-2-hexenal + (E)-2-octenal** (about a 70/30 blend), their 4-oxo derivatives in shed skins, and the aggregation-pheromone components **dimethyl disulfide, dimethyl trisulfide, 2-hexanone and histamine**. We build a pumped e-nose from a **sensor array**: 2x Bosch BME688 running different heater profiles, a Sensirion SGP41, Figaro TGS2602 and TGS2620 MOS sensors, and an optional ION Science MiniPID 2 PPB for extra sensitivity. Air goes in through a "snout" into a sealed chamber, and the sensors between sniffs get a clean-air purge through a carbon filter. A Raspberry Pi 5 runs the sniff cycle and an ML classifier. The nose rides on a Unitree Go2, which maps the room with its built-in LiDAR (ROS 2 + Nav2), stops at waypoints along the bed, baseboards and furniture, and sniffs. When the classifier fires, the robot **sits**, beeps, and drops a pin on the map. A camera plus Hailo AI accelerator adds a second, visual check (bugs, shed skins, fecal spots).

Working as one team

Each robot dog plus its nose is one **unit**, and a handler with the sniff wand is a unit too. A **hub** on the site network hands out duties (patrol, recheck, confirm, calibrate) and collects every sniff and alert. When a dog alerts, the hub automatically sends a *different* unit or a wand handler to confirm, then rolls it all up into a room-by-room report. See [INTEGRATION.md](#).

The most important rule in this plan

Prove the nose before you buy the dog. Phase 0 is a ~\$600 benchtop e-nose. If it can't reliably tell bed-bug training aids apart from blanks and look-alike smells in a blinded test, legs won't fix that. The plan has an explicit go/no-go gate for this.

Questions for reviewers

1. **Robot tier:** Go2 Pro (~\$2.8k, community ROS 2 SDK) or Go2 EDU (\$11k+, official SDK and power dock)?
2. **Phase 0 gate:** are $\geq 85\%$ sensitivity and $\leq 10\%$ false positives at 5 cm the right bar for going ahead with the robot?
3. **Chemicals and live aids:** who holds the account for reference chemicals (Sigma-Aldrich sells mainly to businesses and institutions) and handles live-bug vials?
4. **Ground truth:** which TD-GC-MS lab should we use for the Tenax validation samples?
5. **Field pilot:** which pest-control partner should we approach for Phase 4?
6. **Phase 5:** is there a university entomology or biosensor lab we'd want as a partner?
7. **Hub hosting:** a laptop on site for each job, or one always-on hub reached over a VPN?
8. **Team size:** how many dog units and wand handlers per job (the hub supports any mix)?

Build plan

Prices are approximate as of September 2026 and are there for budgeting. Check each link at checkout. Where a part is sold by many vendors, the link goes to a vendor I found carrying it.

1. Research summary: what we're detecting and what the tech can do

1.1 The bed bug chemical signature

Compound	Where it comes from	Why it matters	Sensor that sees it best
(E)-2-hexenal	Alarm pheromone (~70%), dorsal abdominal glands of nymphs	Strongest, best-studied marker	MOS (BME688, TGS2620, SGP41), PID
(E)-2-octenal	Alarm pheromone (~30%)	Paired with hexenal; the ratio helps tell it apart from plant smells	MOS, PID
4-oxo-(E)-2-hexenal, 4-oxo-(E)-2-octenal	Freshly shed nymph skins (exuviae)	Marker for an active, growing colony	MOS, PID
Dimethyl disulfide / dimethyl trisulfide	Aggregation pheromone (feces and harborages)	Sulfur compounds; MOS sensors respond to them very strongly	TGS2602, BME688
2-hexanone	Aggregation pheromone	Extra dimension for the classifier	MOS, PID
Histamine	Feces, harborages (not volatile)	Not detectable by air; visual/swab only	(camera: fecal spots)
Refuge volatiles: benzaldehyde, hexanal, octanal, nonanal, decanal, (E)-2-nonenal, 1-octen-3-ol, 2-decanone, 3-carene, (3E,5E)-octadien-2-one and more	Papers that bed bugs have lived on (18 compounds the antenna responds to, found by GC-EAG)	These are what a real harborage smells like, beyond the pheromones. But nonanal, decanal and hexanal are also big parts of <i>human skin odor</i> , so a bed smells of them anyway	MOS array (as a pattern), PID

Sources: [Volatile Organic Compounds: A Promising Tool for Bed Bug Detection \(IJERPH 2023, review\)](#), [Heat exposure causes alarm-pheromone emission \(Sci. Reports 2024\)](#), [Oscillator-based trans-2-hexenal sensor array for bed bugs \(Sensors & Actuators B\)](#), [OSU bed bug detection research references](#), [US Patent 10,271,534](#), [Selective detection of bed bug pheromones](#). Aggregation pheromone composition: Gries et al., *Angewandte Chemie* 2015, "Bed bug aggregation pheromone finally identified".

1.2 Hard truths that shape the design

1. **Concentrations are tiny.** A few hidden bugs give off parts-per-trillion to low parts-per-billion levels in room air. A dog's nose works there; a cheap MOS sensor is at its limit. Design responses:
 - **Sniff close.** Put the intake 2 to 10 cm from the surface (the "snout").
 - **Pump and seal.** Pull the sample into a small sealed chamber instead of waiting for it to drift to the sensors.
 - **Measure change, not level.** Purge with carbon-filtered air, then sniff. The classifier reads the *change* from the clean baseline.

- **Optional preconcentrator** (Phase 3): trap VOCs on Tenax TA for 30 to 60 s, then flash-heat to release them all at once. That gives roughly 10x to 100x more signal. [Improving e-noses with a trap + thermal desorption unit](#)
 - **Optional PID** (ION Science MiniPID 2 PPB) as a high-sensitivity channel.
2. **(E)-2-hexenal is also a "green leaf volatile."** Cut grass, fruit, olive oil, cooking and some cleaners give it off. Stink bugs make the same aldehydes. **Negative training data matters as much as positive data.** The sensor array and the hexenal-to-octenal *ratio*, plus the sulfur (DMDS/DMTS) channel, are what give you selectivity.
 3. **Real dogs aren't perfect either.** In a controlled study, dogs found live bugs and eggs with 97.5% accuracy (Pfiester et al. 2008). In real apartments, accuracy dropped a lot: about 44% detection with about 15% false positives (Cooper et al. 2014, Rutgers). That gives us a realistic benchmark to beat and a tested protocol to copy.
 4. **A quadruped can't put its nose into mattress seams.** So the robot sweeps the floor-level zones (bed legs, box-spring base, baseboards, couch base, luggage), and a detachable **sniff wand** on a 1.5 m tube lets a handler check seams and headboards with the same nose.

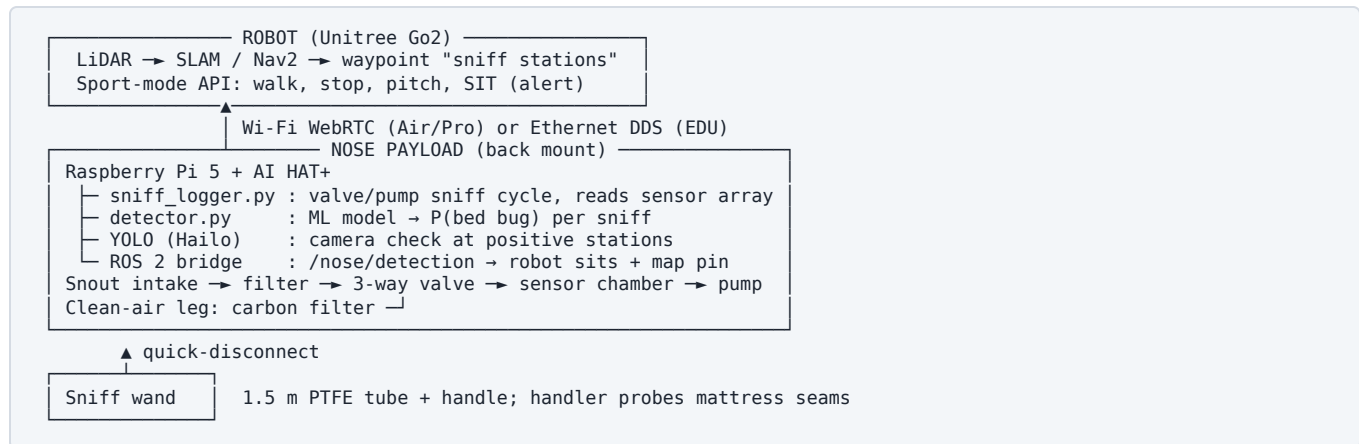
1.3 More research (sources you sent) and what changes because of it

Source	Key point	What it changes in this plan
Knudsen & Ignell 2024, <i>Semiochemicals modulating bed bug behaviour</i> , Curr. Opin. Insect Sci. (ScienceDirect)	Bed bugs are drawn by host odors and heat plus an aggregation pheromone. They have no sex pheromone . Nymphs and adults give off the alarm pheromone to fend off unwanted mating. Their sense of smell uses relatively few odorant receptors	The target list is short and stable, which is good for a small sensor array. The alarm pheromone is released when bugs are disturbed, so the plan adds a "disturb then sniff" step: a short puff or vibration at the spot before sniffing (see Phase 3)
Liu & Liu 2016, <i>Single Sensillum Recording of bed bug olfactory neurons</i> , JoVE 53337, with Liu et al. 2015, <i>Human odorant reception in the bed bug</i> , Sci. Rep. and Liu et al. 2017, <i>Molecular basis of olfactory chemoreception in the bed bug</i>	Bed bug antennae have 6 functional types of smell sensor. Aldehydes and ketones trigger the strongest responses, carboxylic acids trigger almost none. Their odorant receptors have been expressed and tested in lab cells	1) This confirms aldehydes and ketones as the chemical family to tune for: it favours BME688 heater profiles and the PID. 2) It opens a long-term biosensor route (Phase 5). 3) Recording from antennae (SSR/EAG) is how you'd prove which compounds actually matter
Brezolin et al. 2018, <i>Tools for detecting insect semiochemicals: a review</i> , Anal. Bioanal. Chem. (free PDF)	Compares the options: GC with FID/ECD/PID/MS detectors, e-noses, quartz-crystal (QCM), surface-acoustic-wave (SAW), cantilever sensors, and biosensors built on odorant-binding proteins (OBPs) , e.g. an OBP impedance sensor on interdigitated electrodes	GC-MS stays the lab "ground truth" (the Phase 0 validation now includes it). OBP and QCM sensors are the next-generation, high-selectivity sensors to try in Phase 5
Weeks et al. 2020, <i>Semiochemicals from bed bug refuge substrate</i> , Sci. Rep.	18 compounds from harborage paper that bed bug antennae respond to (the table in 1.1)	Adds a harborage class to training: old harborage paper/fabric , not just live bugs. Adds "human-slept bedding" as a required negative, because bedding shares nonanal and decanal
908 Devices ViPiR	A handheld FTIR + Raman analyzer for identifying bulk solids and liquids (drugs, explosives, hazmat). About \$100k+ list price (source)	Not a fit. It identifies a sample you put on it, not trace vapor in air, and it costs 25 times the whole robot. For ground truth, use Tenax tubes sent to a TD-GC-MS lab (typically a few hundred dollars per sample). If you need a portable mass spec later, 908's own MX908 (a handheld mass spec that does trace vapor) is the closer match; ask them for a quote

1.4 Technology options considered

Subsystem	Options	Pick	Why
Gas sensing	MOS (BME688, SGP41, Figaro TGS), PID, QCM/resonant, electrochemical, GC-MS	MOS array + optional PID	MOS is cheap, fast, and each sensor type sees different compounds, so an ML model can combine them. PID adds sensitivity. GC-MS is ground truth but lab-only.
AI on the gas data	Bosch BME AI-Studio, Edge Impulse, scikit-learn on Pi	All three: AI-Studio to design heater profiles, scikit-learn/Edge Impulse for the model	Free, well-documented (BME688 software , Edge Impulse Nicla Sense ME)
Brain	Raspberry Pi 5, Jetson Orin Nano, ESP32	Raspberry Pi 5 (8 GB) + AI HAT+ (26 TOPS)	Cheap, great Python/sensor libraries, runs ROS 2, and the Hailo chip runs YOLO for the visual check
Robot body	Petio Bittle X, WaveShare WAVEGO Pro, Unitree Go2 Air/Pro, Go2 EDU, Boston Dynamics Spot	Unitree Go2 Pro (recommended), Go2 EDU if budget allows	Go2 has built-in 4D LiDAR, a real payload, and ROS 2 support (official on EDU, community go2_ros2_sdk on Air/Pro). Bittle/WAVEGO are fine for demos but can't carry the nose plus a battery reliably. Spot costs more than ten times as much.
Visual check	Pi Camera Module 3, thermal (MLX90640)	Camera Module 3 + YOLO on Hailo	Can spot live bugs, shed skins and fecal spotting close up. Thermal at 32x24 pixels is too coarse for bugs, so it's left out.
Navigation	Go2's built-in LiDAR, add-on RPLIDAR C1	Built-in Go2 LiDAR + slam_toolbox + Nav2	Already on the robot. RPLIDAR C1 only if you use a budget body.

2. System architecture



Detailed drawings are in [the Schematics section](#).

Detection behaviour (copies a detection dog's search)

1. **Map:** tele-op the robot once around the room (slam_toolbox) and save the map.
2. **Plan:** auto-generate "sniff stations" every 30 to 50 cm along the bed perimeter, baseboards, furniture bases, and around luggage.
3. **At each station:** stop, pitch the body nose-down (Go2 Euler / BodyHeight commands), then run 3 sniff cycles (purge 20 s, sniff 5 s, measure 10 s).
4. **Score:** the classifier outputs P(bed bug). Filter it over consecutive sniffs and neighbouring stations.
5. **Alert:** if $P > \text{threshold}$ at 2 or more consecutive sniffs, the robot **sits**, beeps, grabs a camera frame for YOLO, and drops a red pin on the map. Otherwise it moves to the next station.

6. **Report:** a heat-map PNG of P(bed bug) over the room map, plus the CSV log.

3. Phased execution plan

Phase	Weeks	Goal	Spend	Gate to pass
0: Bench nose	1–4	Build the e-nose, collect data, train v1 model	~\$600–800	GO/NO-GO: blinded test with $\geq 85\%$ sensitivity and $\leq 10\%$ false positives at 5 cm, training aid vs. blanks and confounders
1: Sniff wand	5–7	Make it portable (battery, handle, 1.5 m wand), test in a real room	~\$150	$\geq 80\%$ / $\leq 15\%$ in a mock room with hidden vials
2: Robot integration	8–13	Mount the nose on the Go2, set up ROS 2 bridge, stations, SIT alert	Robot + ~\$250	Robot finishes an autonomous room search and alerts on hidden aids
3: Sensitivity upgrades	14–18	Tenax preconcentrator, MiniPID, heater-profile tuning, visual YOLO check	~\$300–1,500	Detects a single live-bug vial at 30 cm; beats the Cooper 2014 field benchmark
4: Field pilot	19–24	Pest-control partner homes, compare against human and dog inspections	Labor	Published accuracy numbers; decide on a product path
5: Next-gen sensor R&D (optional, in parallel)	24+	Biology-based selectivity: OBP/OR biosensor or an insect-antenna sensor	University partner	A sensor channel selective for (E)-2-hexenal/(E)-2-octenal over green-leaf volatiles

Phase 0: Bench e-nose (weeks 1–4)

Build

- Flash Raspberry Pi OS (64-bit, Bookworm or newer) and enable I2C (`sudo raspi-config` → Interface → I2C).
- Wire everything following [Schematics §3](#). Start with the mux and the BME688 pair, check them with `i2cdetect -y 1`, then add one sensor at a time.
- Build the sensor chamber (Schematics §5). Use aluminium, glass or PTFE. **Don't use PLA or PETG for anything the sample air touches:** printed plastics absorb and release VOCs, which smears the signal and carries smells over from one sniff to the next.
- Plumb the pneumatics (Schematics §2). Put the pump **after** the chamber so the pump's own smells never reach the sensors.
- `pip install -r firmware/requirements.txt`, then run `python firmware/sniff_logger.py --label blank -station test`.
- Burn-in:** run the system for 48 hours before you trust any data. New MOS sensors drift a lot at first. Also run the SGP41 conditioning step (the logger does this at startup).

Tune heater profiles (BME688)

- The [BME688 Development Kit](#) (8 sensors) with **BME AI-Studio** can test many heater-temperature profiles at the same time. Record positive and negative samples, let AI-Studio rank the profiles, then run the best two on your two BME688s. This is optional but saves weeks of guessing.

Collect data (with the target presented 5 cm from the snout inlet)

Class	Source	Sessions
bedbug_live	Live-bug training vials	40+

Class	Source	Sessions
bedbug_pseudo	Pheromone pseudo-scent, or a lab mix of (E)-2-hexenal:(E)-2-octenal 70:30 at 1 ppm, 100 ppb and 10 ppb	40+
blank	Clean vial / room air	60+
bedbug_harborage	Paper or fabric strips bed bugs have lived on for 1+ week, sealed in vials (live-bug vendors can supply these, or put a strip in with your own vials)	30+
confounder_*	Bedding a person has slept on for 3+ nights (required) , cut grass, apple, olive oil, cooking oil, cleaners, perfume, dirty laundry, dog, human sweat, stink bug (if available), cockroach aid	10+ each

Ground truth (new): for 10 sessions of each positive class, draw the same air through a Tenax TA tube (200 mL/min × 10 min) and send the tubes to a thermal-desorption GC-MS lab. This confirms the target compounds really reached the snout, and at what concentration, so a miss can be blamed on the sensor and not on the sample.

Train: `python firmware/train_classifier.py data/*.csv`. It does grouped cross-validation (by session, so the model can't cheat by memorising one session) and saves `model.joblib`.

Gate test: someone else sets up 20 blinded trials (10 positive, 10 negative/confounder) in random order. Log the results. **If you miss the gate**, try in this order: move closer, lengthen the sniff, add more negative data, add the MiniPID, add the preconcentrator.

Phase 1: Sniff wand (weeks 5–7)

- Put the Pi + nose in a small enclosure with a 12 V battery (Schematics §4).
- Add a 1.5 m PTFE tube wand with a quick-disconnect at the snout port and a push button on the handle (GPIO 5).
- Mock room: hide vials in a bed frame, a couch and luggage. Run 30 blinded searches.
- This version can already be sold as a pest-control inspection aid.

Phase 2: Robot integration (weeks 8–13)

1. **Set up the robot.** Go2 Pro: install [go2_ros2_sdk](#) (ROS 2 over WebRTC on Wi-Fi). Go2 EDU: use the official [unitree_sdk2 / unitree_sdk2_python](#) over Ethernet DDS ([Go2 SDK guide](#)).
2. **Mount.** Bolt a 3 mm aluminium plate to the Go2's top rails. Nose box on the plate. Snout boom runs forward and down the chest line, intake about 8 cm above the floor at normal stance and 2 to 3 cm when the body pitches down. Keep the total payload under 3 kg for runtime.
3. **ROS 2 bridge node** (Pi 5): publish `/nose/p_bedbug` (Float32) and `/nose/alert` (Bool). A mission node subscribes and sends Nav2 goals to each station. On an alert it calls the sport-mode "Sit" (Pro/EDU) and plays a sound.
4. **Station generator:** load the saved map, take the occupied-cell edges (walls, furniture), offset them 35 cm, and resample every 40 cm into a list of waypoints.
5. **Hub + agent:** run the hub and `firmware/agent.py` on each unit so duties, results and reports go through one place. Link-up schematics for both Go2 models are in [INTEGRATION.md](#).
6. **Test:** 10 runs in the mock room with hidden aids at random locations. Measure detection, false alerts and search time.

Phase 3: Sensitivity upgrades (weeks 14–18)

- **Tenax TA preconcentrator** (Schematics §2b): a stainless tube packed with Tenax TA, wrapped in a polyimide heater with a thermistor. Adsorb for 60 s at 200 mL/min, then heat to about 180 °C in about 5 s and push the released burst into the chamber with a small flow. Sorbent: [Sigma-Aldrich Tenax TA tubes](#) (repack one or cut to length) or [SKC Tenax tubes](#).
- **MiniPID 2 PPB** on ADS1115 channel A2 ([ION Science](#), request a quote).

- **Visual check:** fine-tune YOLOv8n on your own close-up photos (live bugs, shed skins, fecal spots, eggs, and negatives such as lint, seeds and ink dots). Compile for the Hailo-8 with the Hailo Dataflow Compiler. Run it only at positive stations to save power.
- **"Disturb then sniff"** (from Knudsen & Ignell 2024 and the heat-exposure study): bed bugs release alarm pheromone when disturbed. Add a small vibration motor or a 1 s air puff at the station before the SNIFF phase, then compare the sniff with and without it. That tests whether nudging a harborage raises the signal.

Phase 4: Field pilot (weeks 19–24)

- Partner with a licensed pest-control operator. Run the robot **alongside** their normal inspection (visual and/or dog). Confirm every result with a visual find or monitors (for example interceptor cups over 2 weeks).
- Report sensitivity, specificity, time per room, and the cases where it failed.

Phase 5: Next-gen sensor R&D (optional, runs alongside, best with a university partner)

MOS sensors see broad chemical families. Biology gets its selectivity from **receptors**. Three routes, from most to least practical:

1. **OBP impedance biosensor:** coat interdigitated electrodes with an insect odorant-binding protein that binds (E)-2-alkenals and read the impedance change (method in Brezolin 2018 and the [OBP-on-IDE sensor](#)). It reads out through an impedance chip (e.g. AD5933) on the same I2C bus, so it's a drop-in channel for the array.
2. **Coated QCM/SAW:** a quartz crystal microbalance coated with a polymer that attracts aldehydes. It's the same idea as the [oscillator-based trans-2-hexenal sensor array](#).
3. **Cell-based OR sensor or antenna sensor:** express the bed bug odorant receptors (identified by [Liu et al. 2017](#)) in cells, or record from an excised insect antenna (EAG). That's the method the SSR protocol in JoVE 53337 builds on. It's the most sensitive, but it's a research project in its own right: live tissue, a short shelf life, and it needs lab skills.

Use the lab **SSR/EAG** method (JoVE 53337) as a screening tool: it shows which compounds bed bugs themselves respond to, and that helps decide which compounds to calibrate the sensors against.

4. Bill of materials with purchase links

Machine-readable version: [BOM.csv](#).

4.1 Nose: sensing (Phase 0)

#	Part	Qty	~Unit \$	Buy
1	Adafruit BME688 (T/RH/P/gas, AI heater profiles), STEMMA QT	2	20	Adafruit 5046
2	Adafruit SGP41 VOC + NOx MOX sensor, STEMMA QT	1	20	Adafruit 6455
3	Figaro TGS2602 (sulfur compounds, VOC, odor: DMDS/DMTS channel)	1	15	Figaro TGS2602 · Octopart distributors
4	Figaro TGS2620 (alcohols and organic solvents: aldehyde channel)	1	15	Octopart TGS2620
5	Adafruit SHT45 chamber T/RH reference, STEMMA QT	1	13	Adafruit 5665
6	Adafruit ADS1115 16-bit ADC (reads the Figaro and PID outputs)	1	15	Adafruit 1085
7	TCA9548A 8-channel I2C mux, STEMMA QT/Qwiic	1	12	Adafruit 4704
8	STEMMA QT / Qwiic cables, 100 mm, assorted	6	1	Adafruit 4210

#	Part	Qty	~Unit \$	Buy
9	(Optional, speeds up Phase 0) Bosch BME688 Development Kit, 8 sensors	1	87	SparkFun · DigiKey
10	(Optional, Phase 3) ION Science MiniPID 2 PPB	1	quote	ION Science

4.2 Nose: airflow (pneumatics)

#	Part	Qty	~Unit \$	Buy
11	Air pump / vacuum DC motor, 4.5–5 V, 2.5 LPM	2 (1 spare)	8	Adafruit 4699
12	3-way mini solenoid air valve, 12 V, 2-position (NO/NC), 1/8" barbs	1	12–18	Search: "12V 3 way mini solenoid valve air"
13	PTFE tubing 1/8" ID × 1/4" OD, 3 m (sample path, inert)	1	20	McMaster-Carr PTFE tubing
14	Silicone tubing 1/8" ID (pump side and exhaust only)	1	8	McMaster-Carr silicone tubing
15	Inline activated-carbon filter (clean-air purge leg)	2	8	Search: "inline activated carbon air filter 1/8 barb"
16	PTFE membrane particulate filter, 1 µm, barbed (inlet dust guard)	3	5	Search: "PTFE inline filter 1 micron barb"
17	Sensor chamber: 6061 aluminium block 60×40×20 mm (machine it, or use a small aluminium project box) + PTFE gasket	1	15	McMaster aluminium bar · McMaster PTFE sheet
18	Polypropylene barbed fittings/tees, 1/8"	1 pk	10	McMaster barbed fittings
19	Quick-disconnect coupling 1/8" (wand port)	1	8	McMaster quick-disconnects
20	Stainless tube, 1/4" OD × 150 mm (rigid snout)	1	10	McMaster stainless tubing

4.3 Compute, power, control

#	Part	Qty	~Unit \$	Buy
21	Raspberry Pi 5, 8 GB	1	80	raspberrypi.com
22	Raspberry Pi 27 W USB-C PSU (bench only)	1	12	raspberrypi.com
23	Raspberry Pi Active Cooler	1	5	raspberrypi.com
24	microSD 64 GB A2 (or an NVMe HAT + SSD)	1	12	Raspberry Pi SD card
25	Raspberry Pi AI HAT+ 26 TOPS (Hailo-8), Phase 2–3	1	110	raspberrypi.com
26	Raspberry Pi Camera Module 3 (autofocus, close focus)	1	25	raspberrypi.com
27	Pololu D36V50F5 5 V 5.5 A buck regulator (12 V → 5 V rail)	1	25	Pololu 4091
28	IRLB8721 logic-level N-MOSFET (pump, valve, heater drivers)	4	2	Adafruit 355
29	1N5819 Schottky diodes (flyback), 10 kΩ and 100 Ω resistors, 10 kΩ load resistors for the Figaro sensors (1 %)	kit	10	Tayda 1N5819 · resistor kit, any
30	12 V 6 Ah+ Li-ion pack with DC output (Phase 1+), e.g. TalentCell 12 V	1	40	Search: "TalentCell 12V lithium battery"
31	Inline fuse holder + 5 A blade fuse	1	5	any auto parts / Amazon
32	Piezo buzzer (active, 5 V) + 5 mm LED	1	2	Adafruit 1536
33	Momentary push button (wand trigger)	1	1	Adafruit 1009

#	Part	Qty	~Unit \$	Buy
34	Perma-Proto HAT (solder the MOSFET drivers onto it)	1	10	Adafruit 2310
35	Enclosure, aluminium or ABS, ~200×120×75 mm, vented	1	25	Search: "aluminum project enclosure 200x120"

4.4 Robot body (choose one tier)

Tier	Platform	~\$	Buy	Notes
Demo / mascot	Petoi Bittle X	300	Petoi	Too small to carry the nose. Use it for demos with a wand.
Budget	Waveshare WAVEGO Pro (Pi 5 host)	~300–400	CNX writeup / links	Light payload. Needs an external lighter nose (drop the Figaro sensors) and an RPLIDAR C1 for mapping.
Recommended	Unitree Go2 Pro	~2,800	Pricing overview · Robostore	Built-in 4D LiDAR, strong payload. ROS 2 via go2_ros2_sdk (WebRTC).
Pro / research	Unitree Go2 EDU	11,190–15,600	Robostore Go2 EDU · Generation Robots (EDU Plus)	Official SDK and DDS, an expansion dock with power outlets (so no separate battery), onboard Jetson, 12 kg max payload.

4.5 Training aids and reference chemicals

Part	~\$	Buy	Notes
Live bed bug training vials (sealed, mesh-ended)	50–150	Bed Bug Training Aids LLC · Nattaro Labs K9 aids	Bugs survive about 3 months at room temperature. Store in double containment.
Pure-odor aids (no live bugs)	40–100	SOKKS Bed Bug · ScentLogix Bed Bug	Safe to use in client homes
trans-2-Hexenal 98% (E-2-hexenal)	40–60	Sigma-Aldrich 132659	Used to make calibration standards. Many chemical suppliers only sell to businesses or institutions.
trans-2-Octenal	40–70	Sigma-Aldrich search	
Dimethyl disulfide	30	Sigma-Aldrich search	Very strong smell. Use tiny amounts in a fume hood.
Negative / confounder aids: cockroach, other insects	40	Ray Allen K9 training aids	For negative training data

4.6 Budget roll-up

Bucket	Low	High
Phase 0 nose (items 1–8, 11–24, 27–29, 32–34, chemicals, aids)	\$550	\$800
Phase 1 wand/portable (30–31, 35, extra tubing)	\$80	\$150
Phase 2–3 compute and vision (25, 26, Tenax, heater)	\$170	\$300
MiniPID (optional)	\$0	~\$1,000+ (quote)
Robot body	\$300 (WAVEGO)	\$2,800 (Go2 Pro) / \$11–15k (EDU)
Total, recommended path (Go2 Pro, no PID)	≈ \$3,600	≈ \$4,100

5. Software stack

Layer	Tool
OS	Raspberry Pi OS 64-bit + ROS 2 Jazzy (Docker or native Ubuntu 24.04)
Sensor I/O	Adafruit Blinka, <code>adafruit-circuitpython-{tca9548a,bme680,sht4x,ads1x15}</code> , raw I2C for the SGP41, <code>gpiozero</code>
Data + ML	<code>pandas</code> , <code>scikit-learn</code> (RandomForest / gradient boosting), <code>joblib</code> . Optionally Edge Impulse for a microcontroller port
Heater profiles	Bosch BME AI-Studio + BME688 Dev Kit; later port to BSEC2 for on-sensor profiles
Robot	<code>go2_ros2_sdk</code> (Air/Pro) or <code>unitree_sdk2</code> (EDU), <code>slam_toolbox</code> , Nav2
Vision	Ultralytics YOLOv8n → Hailo Dataflow Compiler → <code>hailo-apps</code> / Picamera2

Code in `firmware/`:

- `sniff_logger.py`: runs the purge/sniff/measure cycle, reads all sensors at 2 Hz, writes a CSV with labels.
- `train_classifier.py`: pulls a feature vector out of each sniff, runs grouped cross-validation, saves the model.
- `detector.py`: live inference. Prints P(bed bug), buzzes on an alert, and optionally publishes to ROS 2.

6. Validation protocol (copied from how detection dogs are certified)

1. **Double-blind**: the person who hides the samples isn't in the room during the search. The operator doesn't know how many samples there are or where.
2. **Per-trial result**: a true positive is an alert within 50 cm of the hidden sample. Any other alert is a false positive.
3. **Mix**: each room gets 0, 1 or 2 positives (chosen at random), plus 2 confounders.
4. **Metrics**: sensitivity, specificity, precision, time per room. Report 95% confidence intervals (Wilson).
5. **Benchmarks**: controlled dog studies (~97%) and field dog studies (~44% detection / ~15% false positives, Cooper 2014).
6. **Drift checks**: re-test with the reference vial each day. Recalibrate the baseline if the response drifts more than 20%.

7. Safety, ethics and legal

- **Live bed bugs**: keep sealed vials inside a second sealed container. Never open them in an occupied home. Kill spent vials by freezing at -18 °C for 4+ days. Tell clients and get their consent if you bring live aids on site. Pure-odor aids are the better choice in client homes.
- **Chemicals**: (E)-2-hexenal and 2-octenal are flammable irritants, and DMDS is toxic and very smelly. Read the SDS, wear nitrile gloves and eye protection, and work in a fume hood or outdoors. Make dilutions only in sealed vials.
- **Robot**: Go2 sport mode is strong and fast. Keep people and pets clear during testing, keep the remote e-stop in hand, and cap speed in Nav2. Keep the robot's firmware updated and put it on an **isolated Wi-Fi network**: there have been published security vulnerabilities in consumer robot dogs, including Unitree's BLE setup interface in 2025.
- **Privacy**: the camera only captures at alert stations, and images stay on the device unless the client agrees.
- **Claims**: until Phase 4 data exists, market it as an **inspection aid**, not a certified detector.

8. Stretch ideas

- Mimic dog sniffing: short, fast pump pulses (about 5 Hz) to get more mixing at the inlet.
- Use the same hardware for other targets (mold, termites, gas leaks) by retraining. Only the classifier changes.
- Multi-room fleet reports for hotels (room-by-room heat maps).
- Port the model to an ESP32-S3 / Nicla Sense ME for a \$50 stand-alone "sniff puck" that sits under a bed overnight.

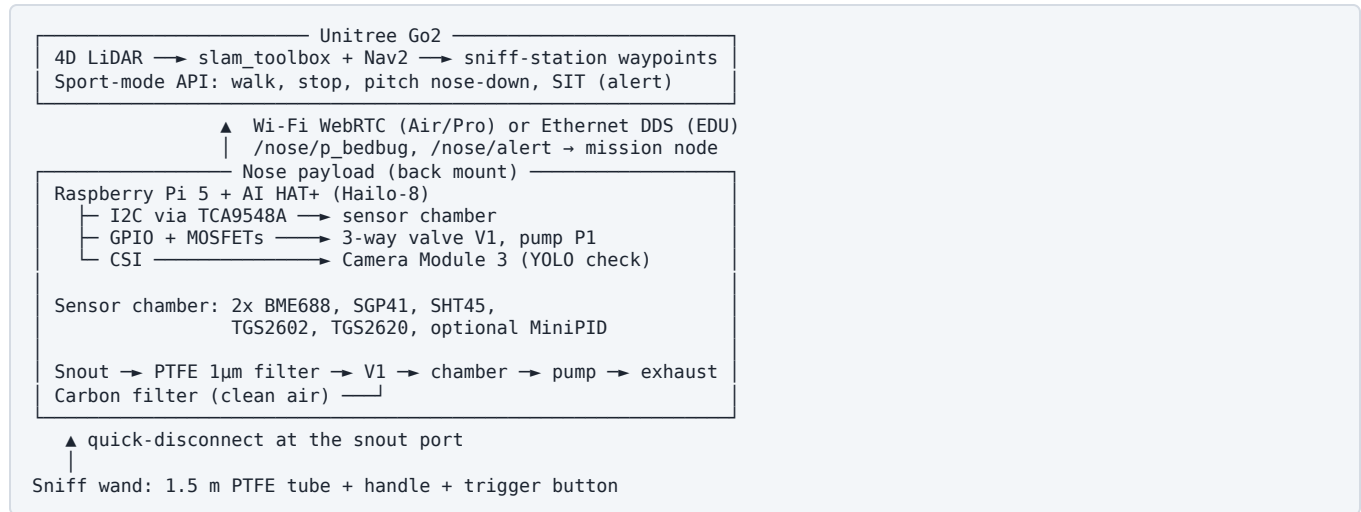
Sources

- [Volatile Organic Compounds: A Promising Tool for Bed Bug Detection](#)
- [Bed bug alarm pheromone emission under heat \(Sci. Rep. 2024\)](#)
- [Oscillator-based trans-2-hexenal sensor array](#)
- [Pfister et al. 2008: dogs detecting live bed bugs and eggs](#)
- [Cooper et al. 2014: field accuracy of bed bug dogs](#)
- [Trap + thermal desorption for e-noses](#)
- [Bosch BME688 · BME688 software](#)
- [Adafruit SGP41 guide](#)
- [Raspberry Pi AI HAT+ docs](#)
- [Unitree Go2 SDK guide · unitree_sdk2_python · go2_ros2_sdk](#)
- [Unitree Go2 EDU buying guide](#)
- [Knudsen & Ignell 2024: Semiochemicals modulating bed bug behaviour](#)
- [Liu & Liu 2016: Single sensillum recording in bed bugs \(JoVE 53337\)](#)
- [Liu et al. 2015: Human odorant reception in the bed bug · Liu et al. 2017: bed bug odorant receptors](#)
- [Brezolin et al. 2018: Tools for detecting insect semiochemicals](#)
- [Weeks et al. 2020: semiochemicals from bed bug refuge substrate](#)
- [908 Devices VipIR \(evaluated, not recommended\)](#)

Schematics

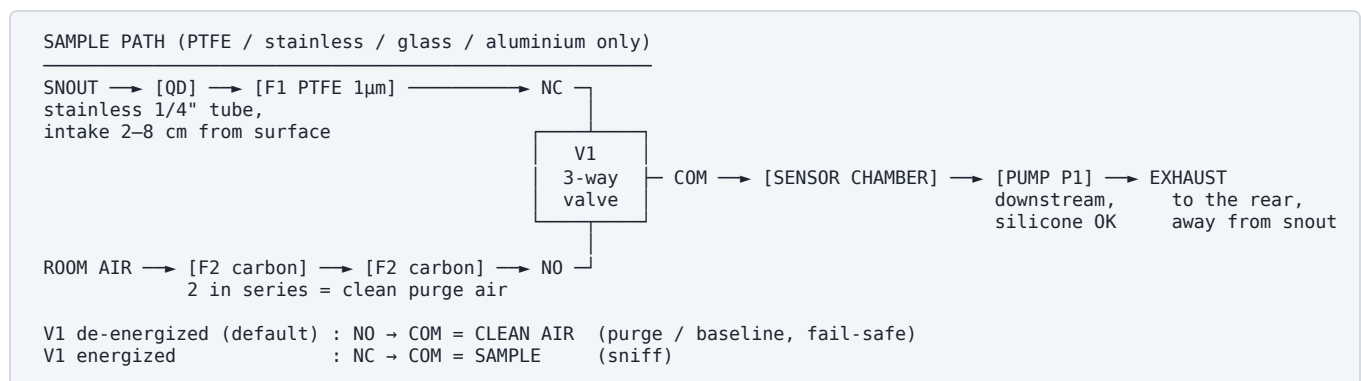
These cover the system, the airflow (pneumatics), the electronics, power, and the mechanical parts. Part numbers (#) refer to the BOM in [Build plan §4](#).

1. System block diagram



2. Airflow (pneumatic) schematic

2a. Base system (Phases 0–2)



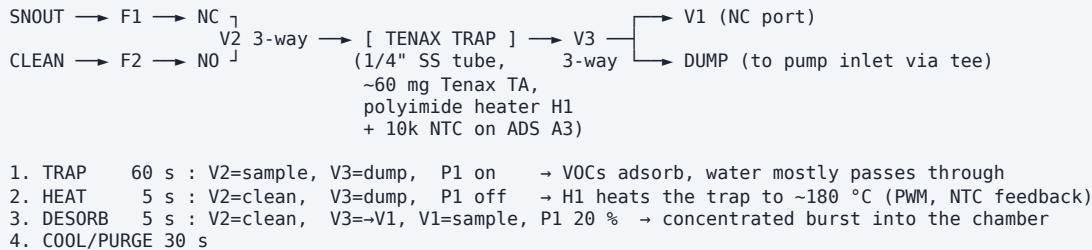
Sniff cycle (set in firmware/sniff_logger.py):

Step	Time	V1	P1	What the sensors see
PURGE	20 s	off (clean)	100 %	Carbon-filtered air → baseline R0
SNIFF	5 s	on (sample)	100 %	Sample fills the chamber (~0.5 s exchange)
HOLD	10 s	off	0 %	Static sample. MOS response peaks, heater profiles sweep
(repeat)				Features = response vs. baseline, per sensor

Rules for the sample path:

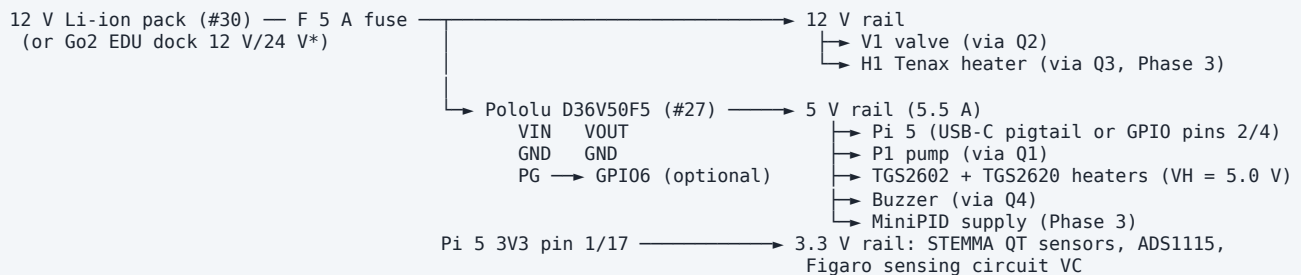
- Everything from the snout to the chamber must be PTFE, stainless steel, glass or aluminium.
- Keep the tube from the snout to the chamber under 40 cm, or 1.5 m on the wand. Shorter means less stickiness and faster response.
- The pump goes **downstream** of the chamber. Its exhaust points backward so the robot doesn't re-sniff it.

2b. Phase 3 add-on: Tenax TA preconcentrator



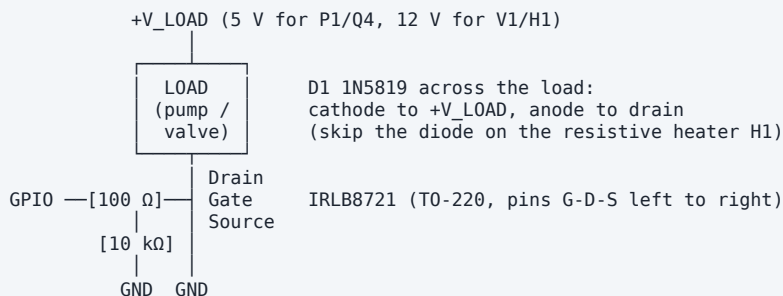
3. Electrical schematic

3a. Power tree



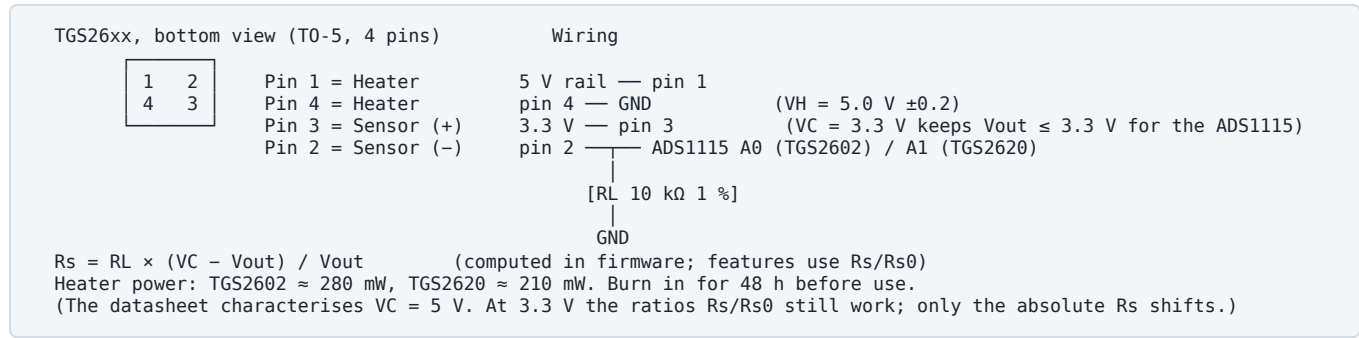
* On the Go2 EDU dock the D36V50F5 accepts up to 50 V in, so it can run from the 24 V outlet.
 If you power the Pi 5 through the GPIO 5 V pins, add `usb_max_current_enable=1` to `/boot/firmware/config.txt`.
 All grounds are tied at one star point on the Perma-Proto HAT.

3b. Low-side MOSFET drivers (x4, IRLB8721 #28), all the same

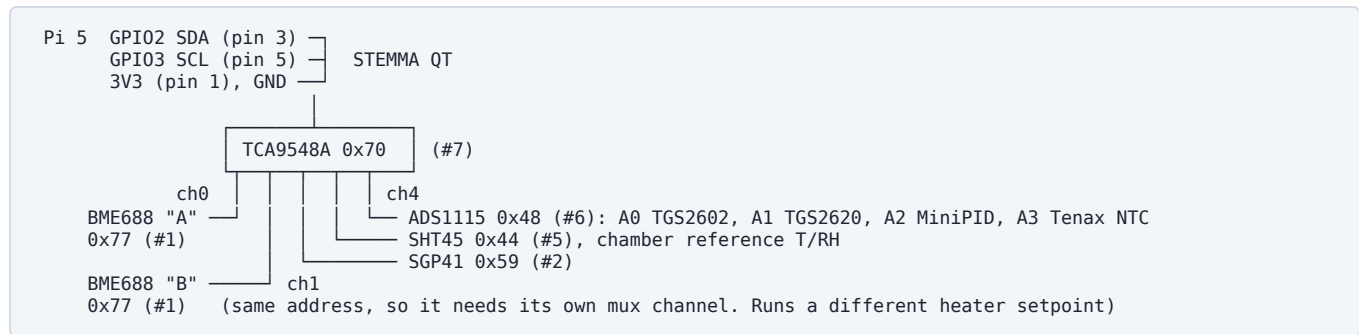


Q1: GPIO18 (pin 12, PWM) → P1 pump 5 V, ~0.5 A
 Q2: GPIO23 (pin 16) → V1 3-way valve 12 V, ~0.2 A
 Q3: GPIO24 (pin 18, PWM) → H1 Tenax heater 12 V, ~1 A (Phase 3)
 Q4: GPIO25 (pin 22) → Buzzer (active) 5 V, ~30 mA
 The 10 kΩ pull-down runs from gate to GND so the load stays off while the Pi boots.

3c. Figaro TGS2602 / TGS2620 circuit (x2, the same)



3d. I2C tree



3e. Raspberry Pi 5 header map

Pin	GPIO	Function	Connects to
1	3V3	3.3 V out	STEMMA QT V+, Figaro VC, NTC divider
2, 4	5V	5 V in (optional)	D36V50F5 VOUT
3	GPIO2	I2C1 SDA	TCA9548A SDA
5	GPIO3	I2C1 SCL	TCA9548A SCL
6, 9, 14...	GND	Ground	Star ground
12	GPIO18	PWM0	Q1 gate → pump P1
16	GPIO23	Output	Q2 gate → valve V1
18	GPIO24	PWM (software)	Q3 gate → Tenax heater H1
22	GPIO25	Output	Q4 gate → buzzer
29	GPIO5	Input, pull-up	Wand trigger button → GND
31	GPIO6	Input	Buck regulator PG (optional)
36	GPIO16	Output	Alert LED (330 Ω → LED → GND)
CAM0	CSI	Camera	Camera Module 3 (#26)
PCIe	FFC	PCIe x1	AI HAT+ (#25)

Stack order: Pi 5 → Active Cooler → AI HAT+ (with its stacking header) → Perma-Proto HAT (MOSFET drivers, Figaro circuits, connectors).

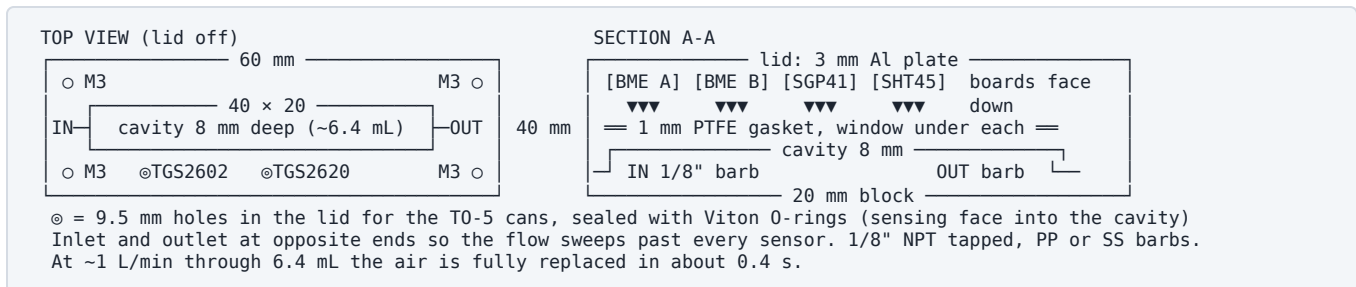
4. Power budget

Load	Rail	Avg W	Peak W
Pi 5 (active, ROS 2 + ML)	5 V	5.0	8.0
AI HAT+ (only runs at alert stations)	5 V	0.5	2.5
Camera Module 3	5 V	0.3	1.0
Pump P1 (~70 % duty)	5 V	1.8	2.5
TGS2602 + TGS2620 heaters	5 V	0.5	0.5
BME688 ×2, SGP41, SHT45, ADS1115, mux	3.3 V	0.1	0.2
Valve V1 (~15 % duty)	12 V	0.4	2.4
Tenax heater H1 (Phase 3, ~10 % duty)	12 V	1.2	12
Buck efficiency loss (~90 %)	–	0.9	1.5
Total		≈ 10.7 W	≈ 30 W

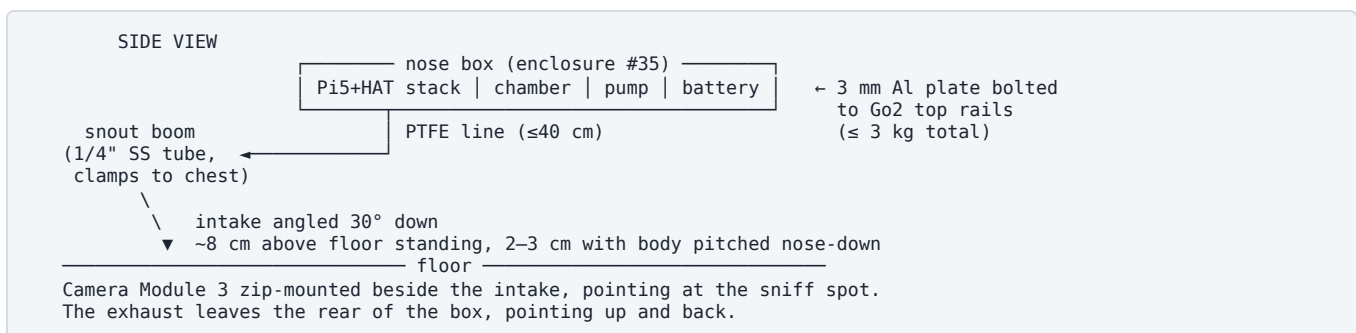
A 12 V 6 Ah pack (72 Wh) gives about **5–6 hours**, which is longer than the Go2's own 1–4 hour runtime.

5. Mechanical

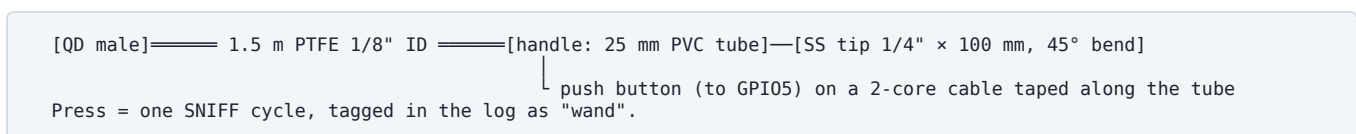
5a. Sensor chamber (6061 aluminium, #17)



5b. Snout and mount on the Go2



5c. Sniff wand (Phase 1+)



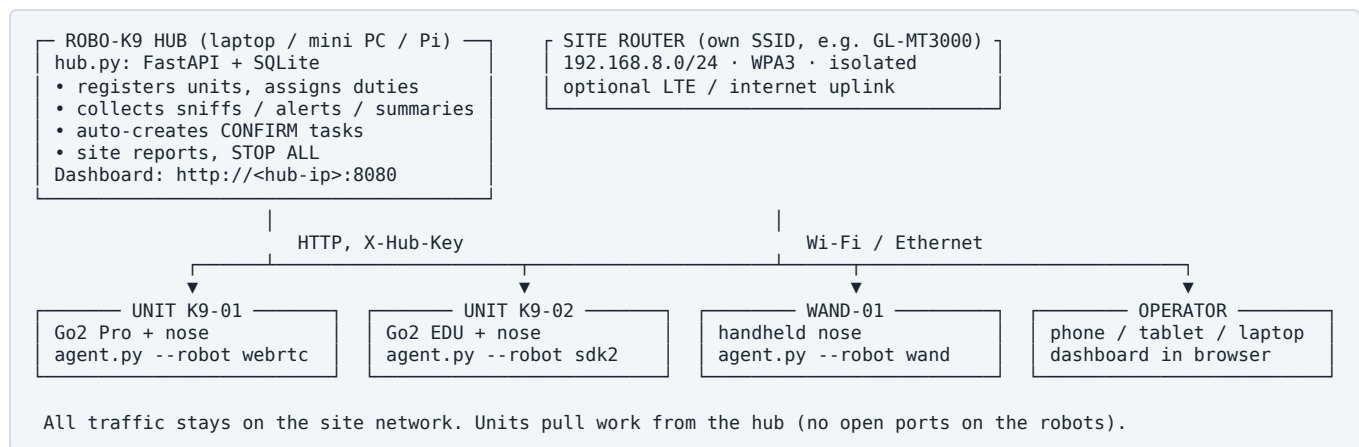
Robot dog link-up and hub

This document covers how the e-nose payload **links up with an existing robot dog** (Unitree Go2) so the two work as one unit, and how several units, wand handlers and operators **work as one team** through a central hub that assigns duties, collects results and writes reports.

Yes, the plan uses an off-the-shelf robot dog. Nothing on the robot is modified. The nose payload bolts onto the top of the robot, talks to it through Unitree's own interfaces, and commands it the way any app would.

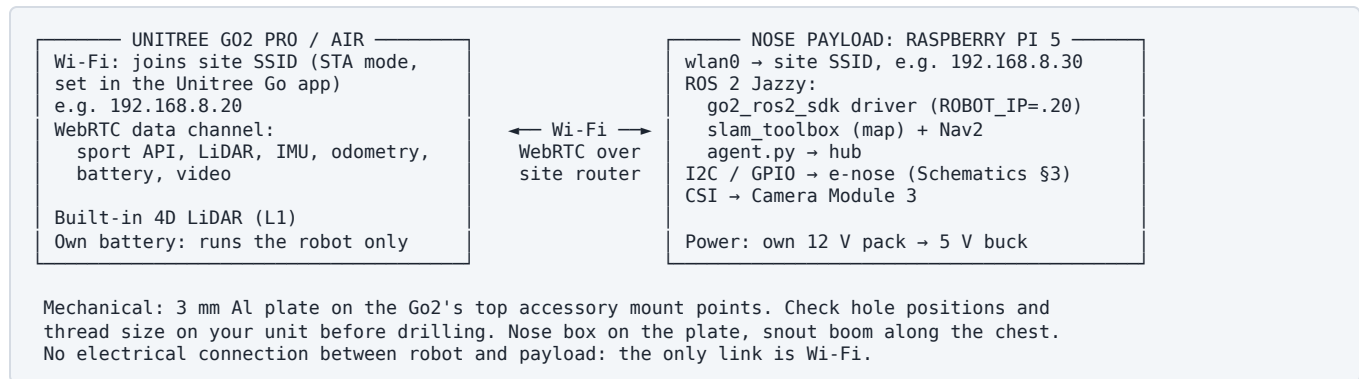
Piece	Runs on	Code
Unit = robot dog + nose payload	Raspberry Pi 5 in the nose box	firmware/agent.py , firmware/robot.py
Wand unit = a handler with the handheld nose	Raspberry Pi 5 in the wand pack	same agent, --robot wand
Hub = assigns duties, collects data, reports	Any laptop, mini PC or spare Pi on site	hub/hub.py , dashboard hub/static/index.html

1. Team overview (hub + units)



Why the units pull instead of the hub pushing: robots move between Wi-Fi cells and drop out. With the pull model, a unit that loses Wi-Fi just catches up when it reconnects, and nothing on the robot has to accept incoming connections. If a unit goes silent for 60 s, the hub puts its task back in the queue for another unit.

2. Link-up schematic: Go2 Pro / Air (Wi-Fi)

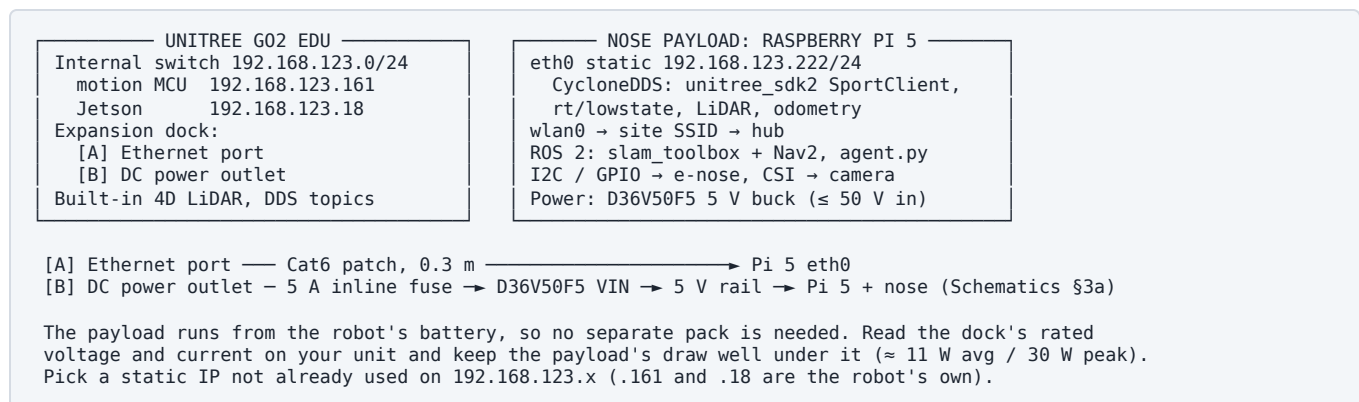


Steps

1. In the Unitree Go app, switch the Go2 to **STA mode** and join the site router's SSID. Give it a DHCP reservation (e.g. 192.168.8.20).
2. On the Pi: install ROS 2 Jazzy and [go2_ros2_sdk](#). Set `ROBOT_IP` to the Go2's address and the connection type to WebRTC, as its README describes, then launch the driver. It publishes LiDAR, odometry, joint states and battery, and takes sport commands on `/webrtc_req`.
3. Build the room map once with `slam_toolbox` (tele-op the robot around the room), save it, and run Nav2 on that map.
4. Start the agent: `python agent.py --hub http://<hub-ip>:8080 --key $HUB_KEY --unit-id k9-01 --robot webrtc --model model.joblib --stations stations.json`.

Power: the Air/Pro has no official accessory power outlet, so the payload carries its own 12 V pack (Schematics §3a). Don't tap the robot's battery.

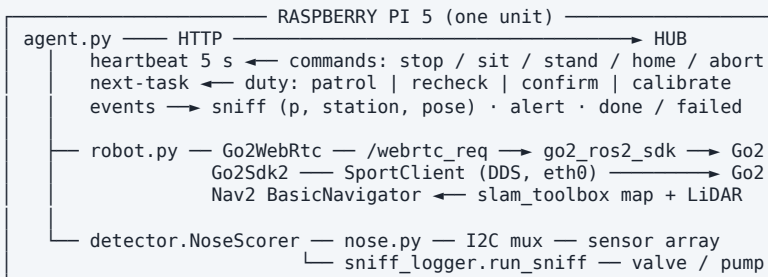
3. Link-up schematic: Go2 EDU (Ethernet + dock power)



Steps

1. Plug the Pi's Ethernet into the EDU expansion dock and give `eth0` a static address on 192.168.123.0/24.
2. Install [unitree_sdk2_python](#) (CycloneDDS). Check it works with the SDK's examples before adding the nose.
3. Map and navigate as on the Pro (`slam_toolbox` + Nav2 on the Pi, or on the EDU's own Jetson).
4. Start the agent with `--robot sdk2 --robot-model "Unitree Go2 EDU"`.

4. Software link-up inside one unit



Robot action	Go2 Pro/Air (Go2WebRtc)	Go2 EDU (Go2Sdk2)
Walk to a sniff station	Nav2 goToPose	Nav2 goToPose
Nose-down sniff posture	sport StopMove + Euler(pitch 0.30 rad)	SportClient.StopMove() + Euler()
Alert = SIT	sport StopMove + Sit	SportClient.Sit()
Stand back up	RiseSit + BalanceStand	RiseSit() + BalanceStand()
Emergency stop	Nav2 cancel + StopMove	Nav2 cancel + StopMove()

Sport-mode API ids live in one table, `SPORT_API` in `robot.py`. **Check them against the Go2 firmware and SDK versions you install**, because Unitree has changed them between releases. The robot's own remote-control e-stop always overrides everything here.

5. The hub: duties, data and reports

Robo-K9 Hub

Hub key

Updated 9:49:11 PM

STOP ALL UNITS

Units

Unit	State	Battery	Task	Last seen	
K9-01 (Go2 Pro) Unitree Go2 Pro	idle	100%		7 s ago	Sit Home
K9-02 (Go2 EDU) Unitree Go2 EDU	idle	100%		7 s ago	Sit Home
Wand-01 (handler) Handheld sniff wand	idle	100%		7 s ago	Sit Home

Assign a duty

patrol: search a room
Site (e.g. Hotel 12)

Room (e.g. 204)
Normal

Any capable unit
Queue task

Any alert automatically creates an urgent **confirm** task for a different unit or a wand handler.

Tasks

Task	Kind	Site / room	Status	Unit
T-125e6ffd	recheck	Hotel 12 / 205	queued	
T-ddb6f42e	confirm	Hotel 12 / 204	done	wand-01
T-13b80dec	patrol	Hotel 12 / 205	done	k9-01
T-c5b7a628	patrol	Hotel 12 / 204	done	k9-02

Alerts

When	Site / room	Unit	P (bed bug)	Station	Confirmation
9/25/2026, 9:48:53 PM	Hotel 12 / 204	k9-02	0.89	4	ACTIVE confirmed
9/25/2026, 9:48:50 PM	Hotel 12 / 204	k9-02	0.83	3	ACTIVE confirmed

Site report

Site
Build report

Download JSON

Room	Verdict	Searches	Sniffs	Alerts	Confirmed	Max P
------	---------	----------	--------	--------	-----------	-------

Screenshot of the hub running a full simulation: two robot units and one wand unit, fake bed bug hot spot in room 204.

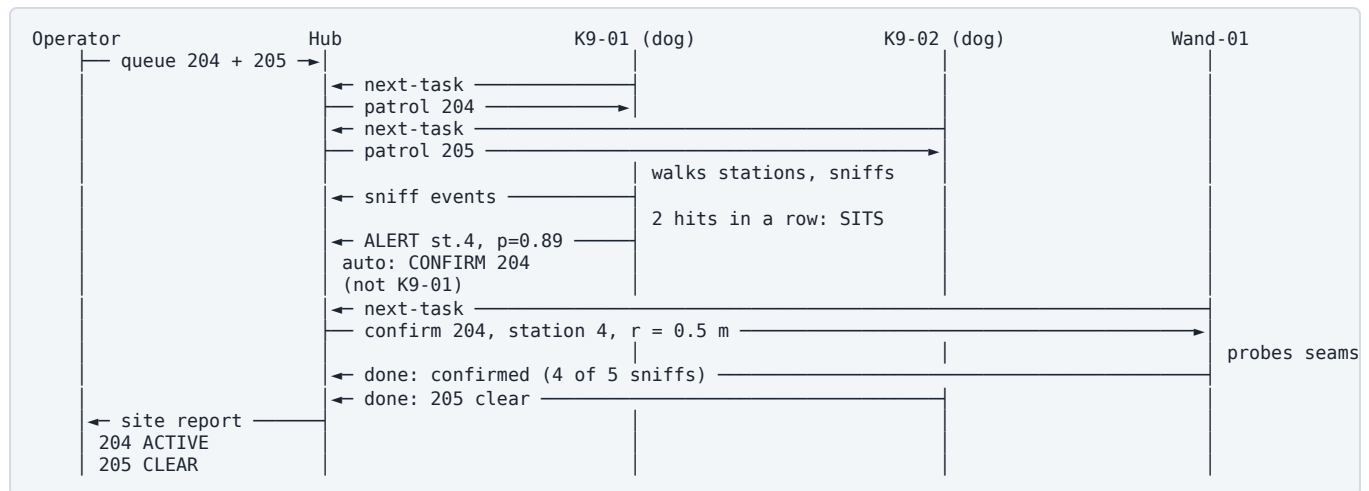
5.1 Duties the hub can assign

Duty	Who can take it	What the unit does	What it reports
patrol	Units with <code>patrol</code> (the dogs)	Walks every sniff station in the room, 3 sniffs each. Sits after 2 hits in a row	Every sniff (P, station, pose), each alert, a summary
recheck	same	Same as patrol, used for follow-up after treatment	same
confirm	Units with <code>confirm</code> (dogs or wands)	Goes to the alert spot and takes 5 sniffs. Confirmed if ≥ 2 are over the threshold	confirmed yes/no, all P values
calibrate	Units with <code>nose</code>	Reference vial $\times 3$, blank $\times 3$	pass/fail, P values (drift check)

5.2 Rules the hub applies automatically

- **Second opinion:** an alert on a patrol creates an **urgent confirm task** for a *different* unit or a wand handler, never the unit that alerted. Alerts within 0.5 m of an open confirm task share it.
- **Right unit for the job:** each duty needs a capability. A wand never gets sent on a patrol, and you can also target one unit by name.
- **Priority:** urgent (1) before normal (3) before low (5), then oldest first.
- **Nothing gets lost:** a unit silent for 60 s is marked offline and its running task goes back in the queue.
- **One task, one unit:** tasks are claimed in a locked transaction, so two units polling at the same moment can't both take it.
- **Operator control:** Stop all, per-unit Sit / Home, and cancel a task (which aborts it on the unit). Commands ride on the next heartbeat reply, so they arrive within about 5 s.

5.3 A mission, start to finish



5.4 Room verdicts in the site report

Verdict	Meaning
ACTIVE - confirmed	At least one alert was confirmed by a second unit or a wand
SUSPECT - confirmation pending	Alert raised, confirm task still queued or running
SUSPECT - alert with no confirmation	Alert raised but no confirm was run (e.g. it was cancelled)
CLEAR - alerts not confirmed	Alerts happened but the second opinion didn't back them up
CLEAR	Searched, no alerts
NOT SEARCHED	Task queued or failed (e.g. room not mapped yet)

The report can be downloaded as JSON from the dashboard. It lists, per room: searches, sniffs, alerts, confirmations, the highest P(bed bug) seen, and which units took part.

5.5 Hub API (all calls need the `X-Hub-Key` header)

Method + path	Used by	Purpose
POST <code>/api/units/register</code>	unit	Join the team with capabilities
POST <code>/api/units/{id}/heartbeat</code>	unit	State, battery, pose → gets back any pending command
POST <code>/api/units/{id}/next-task</code>	unit	Claim the most urgent task it can do
POST <code>/api/tasks/{id}/events</code>	unit	progress / sniff / alert / done / failed
POST <code>/api/tasks</code>	operator	Queue a duty {kind, site, room, priority, target_unit, params}
POST <code>/api/tasks/{id}/cancel</code>	operator	Cancel (aborts it on the unit if running)
POST <code>/api/units/{id}/command</code>	operator	stop, sit, stand, return_home, abort_task
POST <code>/api/stop-all</code>	operator	Stop every unit
GET <code>/api/units, /api/tasks, /api/alerts, /api/reports/{site}</code>	dashboard	Live views and reports

6. Network and security

- Run the team on its **own SSID / VLAN**, and don't put robots on a client's guest Wi-Fi. A travel router like the GL.iNet GL-MT3000 is enough.
- Set a long random `HUB_KEY`. Every unit and the dashboard use it.
- For remote viewing, put the hub behind a VPN (e.g. Tailscale or WireGuard) rather than opening a port to the internet.
- Keep the Go2 firmware updated and isolated. Consumer robot dogs have had published vulnerabilities (BUILD_PLAN §7).
- The hub stores only sniff scores, poses and summaries, no camera images, so client privacy is simple to explain.

7. Extra parts for the link-up

Part	Qty	~\$	Buy	Notes
Travel router, e.g. GL.iNet GL-MT3000 (Beryl AX)	1 per team	90	GL.iNet	Private site network for hub + units
Hub computer: any laptop, or a Raspberry Pi 5 4 GB + case + SSD	1 per team	0–120	raspberrypi.com	Runs <code>hub.py</code> . Very light load
Cat6 patch cable, 0.3 m (EDU only)	1 per EDU unit	5	any	Pi eth0 → dock
Fused DC lead to dock connector (EDU only)	1 per EDU unit	10	match your dock's connector	5 A inline fuse
Tablet or phone for the operator	1	own	–	Dashboard runs in any browser

8. Try it with no hardware (simulation)

```
# terminal 1: hub
cd robot-scent-dog/hub && pip install -r requirements.txt
HUB_KEY=demo uvicorn hub:app --port 8080
# terminals 2-4: two simulated dogs and a simulated wand, bed bug hot spot at (2.0, 1.0)
cd robot-scent-dog/firmware
python agent.py --hub http://localhost:8080 --key demo --unit-id k9-01 \
  --robot sim --nose sim --sim-hotspot 2.0,1.0 --stations stations.example.json
python agent.py --hub http://localhost:8080 --key demo --unit-id k9-02 \
  --robot sim --nose sim --sim-hotspot 2.0,1.0 --stations stations.example.json
python agent.py --hub http://localhost:8080 --key demo --unit-id wand-01 \
  --robot wand --caps confirm,nose --nose sim --sim-hotspot 2.0,1.0
# open http://localhost:8080, enter key "demo", queue patrols for site "Hotel 12", rooms 204 and 205
```

Result of this simulation: the two dogs split the rooms. The dog that got room 204 alerted twice near the hot spot and sat. The hub opened one confirm task and sent it to the wand, which confirmed it. The report came back **204 ACTIVE - confirmed, 205 CLEAR**. A room with no mapped stations failed cleanly and was marked NOT SEARCHED. Hub tests: `cd hub && pytest -q` (9 tests: claiming, auto-confirm, capabilities, requeue, stop-all, verdicts).